

Functional Interaction Transfer from Human to Robot

We do not transfer human actions.
We transfer the functional interaction realized by human actions,
and let each robot embodiment synthesize its own actions to realize the same interaction.

Human Experience → Shared Task Knowledge → Robot Action

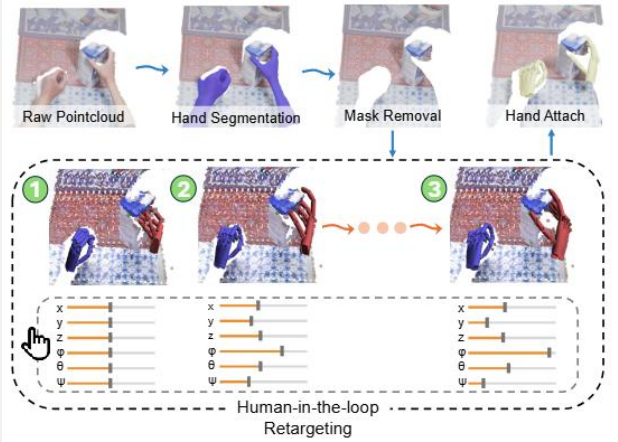
A: What to be extracted?

B: How to transfer across embodiments?

- 
- ① Action
 - ② Effect
 - ③ Interaction

Action

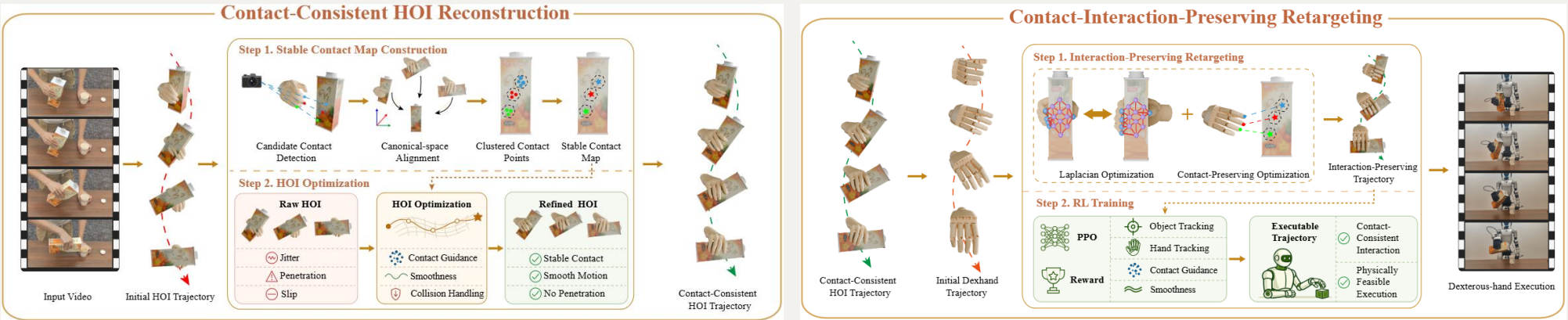
• [UniDex](#) Human-in-the-Loop retargeting



• [AdvDex](#) Joint-aligned Action Space

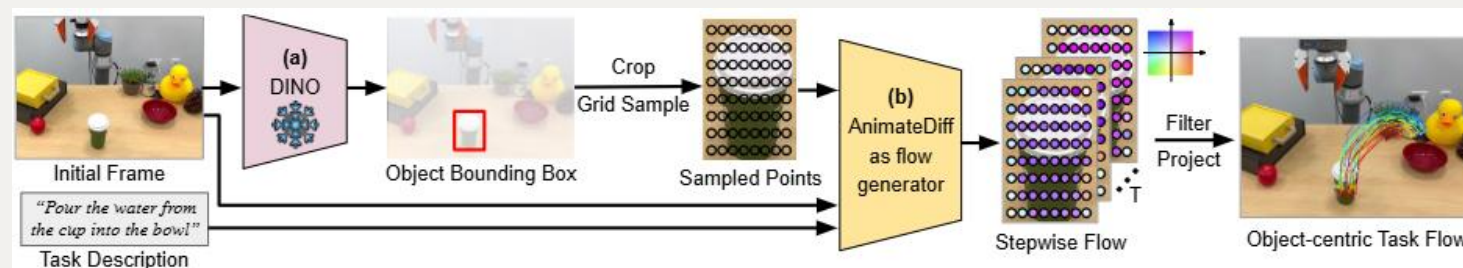


• [C2Dex](#) Human Video $\rightarrow$ 恢复hand-object trajectory $\rightarrow$ 稳定object-side contacts $\rightarrow$ 接触约束retarget

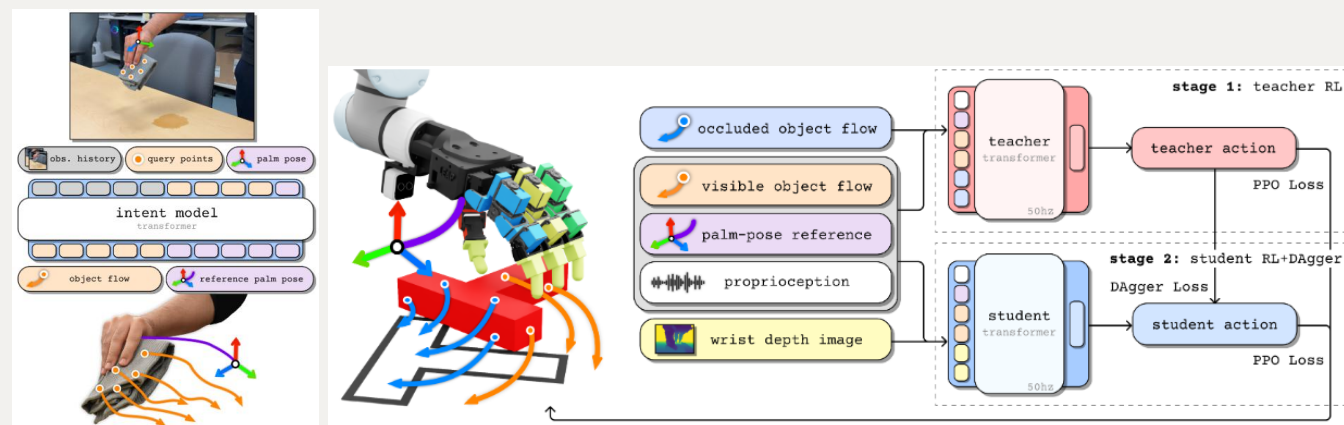


Effect

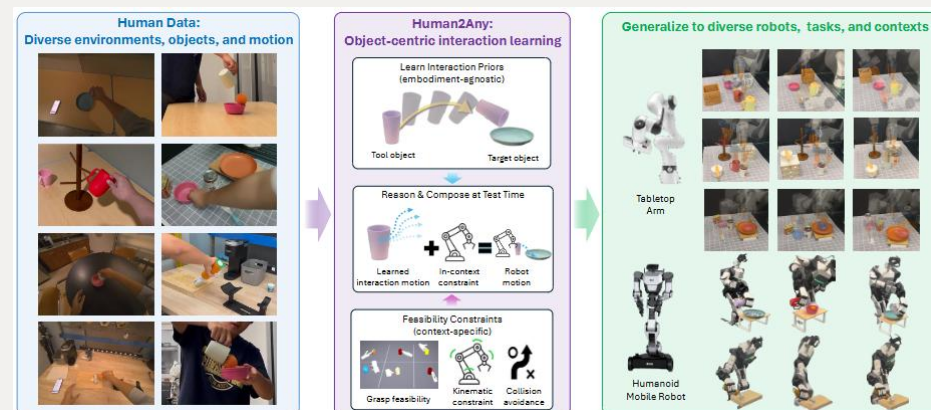
- [Im2Flow2Act](#) Human Video: scene $\rightarrow$ object flow ; Robot Sim Play Data: desired flow $\rightarrow$ robot action



- [LUCID](#) Object flow + Palm pose



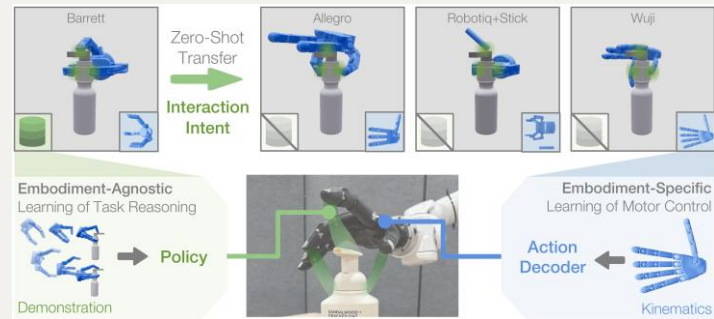
- [Human2Any](#) Tool & Target Object relative motion + constraint rules



Interaction

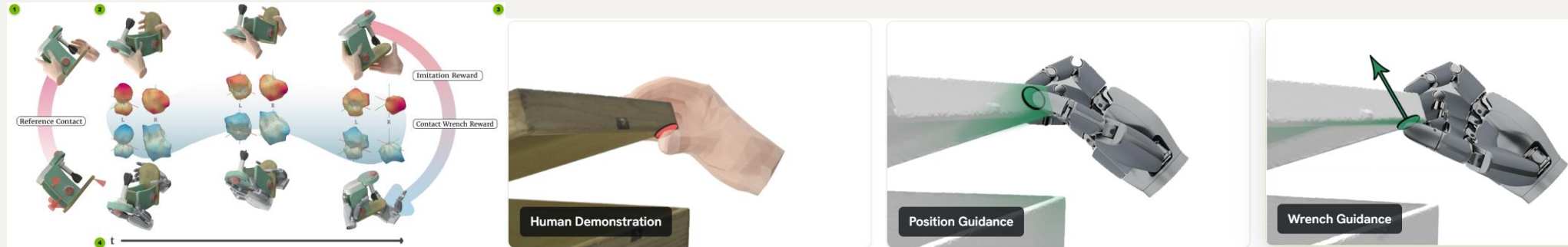
•[KITE](#)

Spatial Engaging Area + Direction $\rightarrow$ Policy ; URDF $\rightarrow$ interaction intent $\rightarrow$ Decoder



•[CHORD](#)

Contact Wrench Space (each task needs a specialized RL Policy)



•[ContactFlow](#)

3D Contact Point Trajectory $\rightarrow$ World Model $\rightarrow$ Future Outcome



Human Experience → Shared Task Knowledge → Robot Action

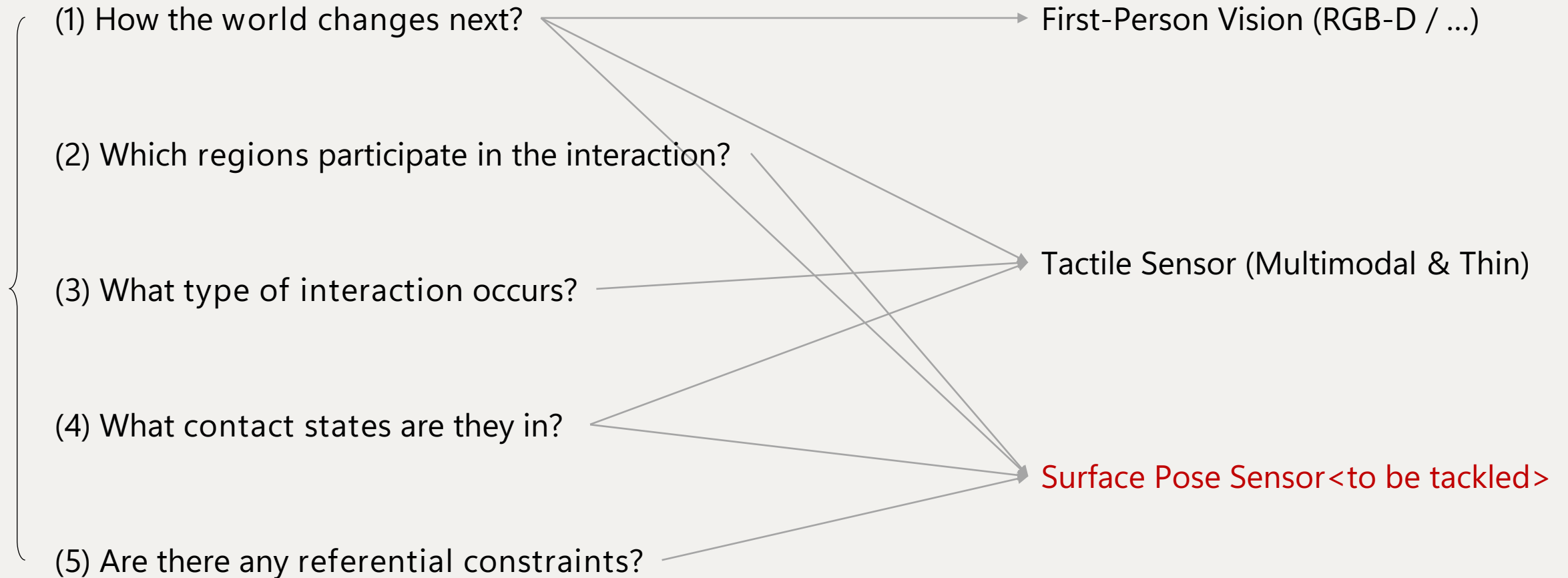
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A: What to be extracted?

I . Human Interaction Acquisition



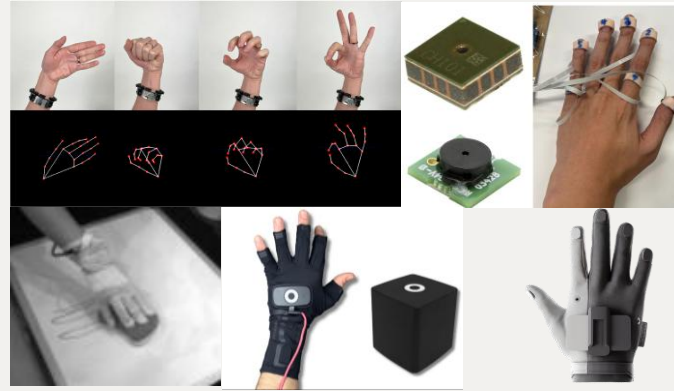
No need for human hand poses / skeleton, just focus on interaction surfaces

A: What to be extracted?

I . Human Interaction Acquisition

Outer Field

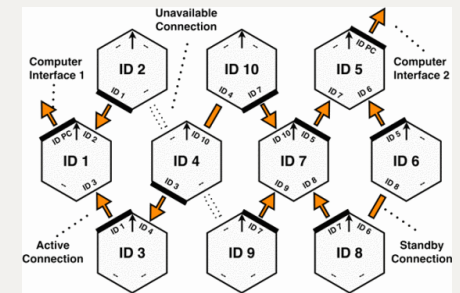
Pose Only



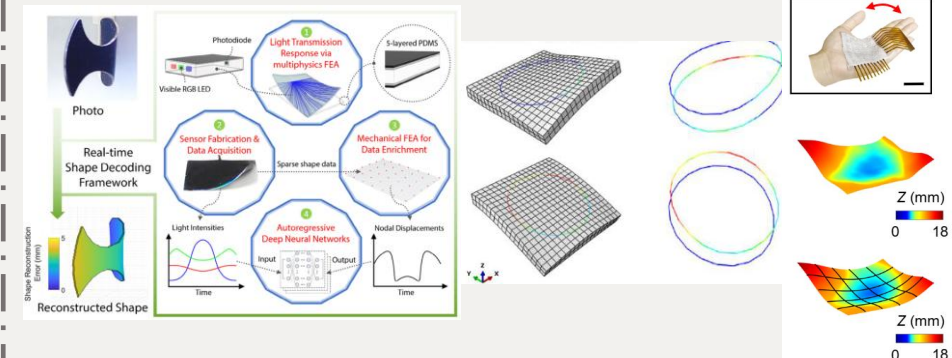
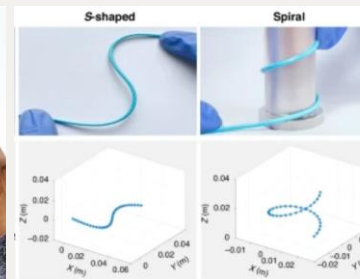
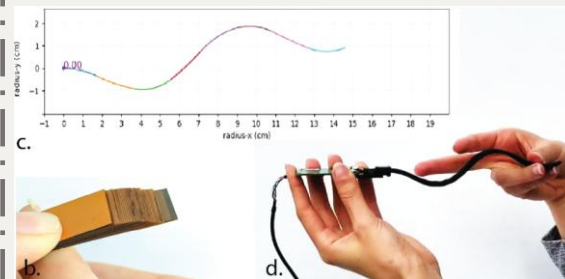
Pose & Geometry



Mechanical Constraint



Self-Aware Soft Constraint



Different people do the same thing differently. So we need to find the shared task structure.

We train a model that predicts the future interaction with a universal representation.

(1) How the **world changes** next?

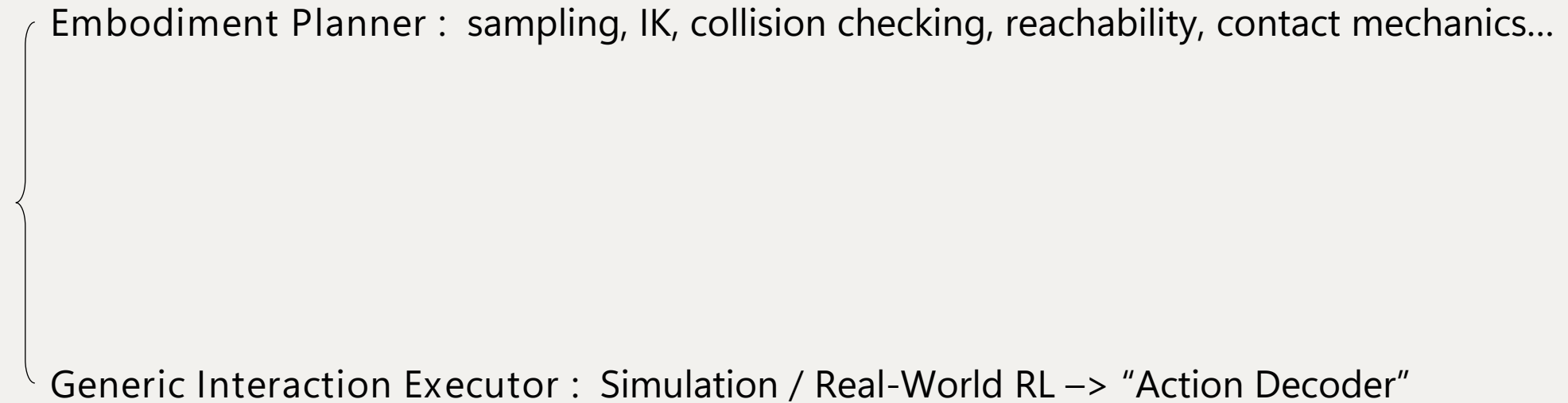
(2) Which **regions** participate in the interaction?

(3) What **type of interaction** occurs?

(4) What **contact states** are they in?

(5) Are there any **referential constraints**?

We build a generic embodiment executor to adapt diverse end-effectors.



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